

Task No.: 6**Date: 08-01-2026**

Title: Develop an Android application to execute Threads and Multithreads**Aim:**

To develop an Android Application that implements Multithreading.

Code:*MainActivity.kt*

```
package com.example.madtask7
```

```
import android.os.Bundle
import android.widget.Button
import android.widget.ImageView
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {
```

```
    lateinit var imageView: ImageView
    lateinit var button1: Button
    lateinit var button2: Button
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
```

```
        imageView = findViewById(R.id.imageView)
        button1 = findViewById(R.id.button1)
        button2 = findViewById(R.id.button2)
```

```
        button1.setOnClickListener {
            Thread {
                imageView.post {
                    imageView.setImageResource(R.drawable.image1)
                }
            }.start()
        }
    }
```

```

button2.setOnClickListener {
    Thread {
        imageView.post {
            imageView.setImageResource(R.drawable.image2)
        }
    }.start()
}
}
}

```

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <!-- Top Center Section -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:gravity="center_horizontal"
        android:padding="16dp">

        <TextView
            android:id="@+id/txtName"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Name: Yoga Shri P K"
            android:textSize="18sp"
            android:textStyle="bold" />

        <TextView
            android:id="@+id/txtRoll"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Roll No: 23BCS192"
            android:textSize="18sp"
            android:textStyle="bold"

```

```
        android:layout_marginTop="4dp" />
</LinearLayout>
```

```
<!-- Center Content -->
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="365dp"
        android:layout_height="344dp"
        android:layout_margin="20dp"
        android:contentDescription="Image Display"
        android:src="@mipmap/ic_launcher" />

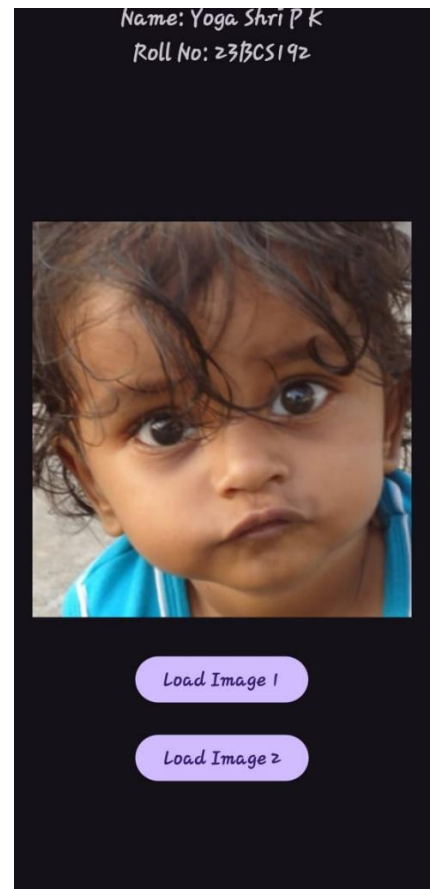
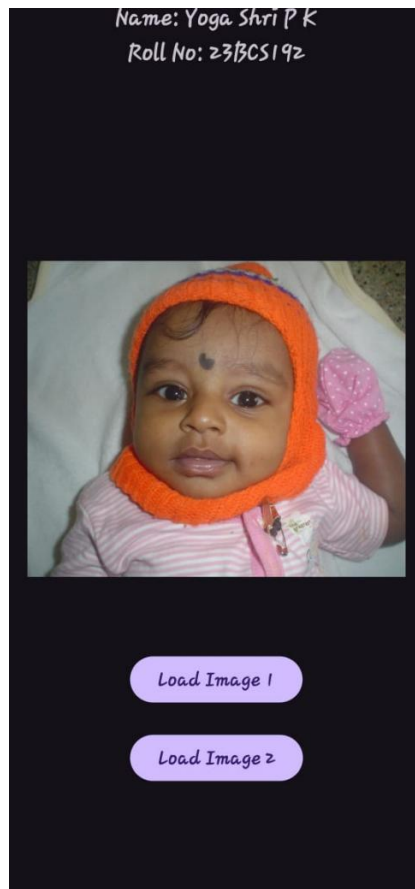
    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Load Image 1"
        android:layout_margin="10dp" />

    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Load Image 2"
        android:layout_margin="10dp" />

</LinearLayout>
```

```
</LinearLayout>
```

Output:



Result:

Thus, Android Application that implements multi-threading is developed and executed successfully.